BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 24

Note: Note: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Mark.

Q1. 1. Choose the correct Answer for each from the given question:

Q1. Science of identification and classification of organism is:

(A) Histology
(B) Cytology
(C) Taxonomy
(D) Anatomy

Q2. Scientific name of frog is:

(A) Palaeon
(B) Rana tigrina
(C) Periplaneta
(D) Pheretima

Q3. Volvox is a polyphyletic genus which belongs to:

(A) Green algae
(B) Red algae
(C) Brown algae
(D) None of these

Q4. Biological method involves all of the following, except :

(A) Data collection
(B) Observation
(C) Experiment
(D) Proportion

Q5. Field of research that examines mathematical representation of : biological system, is called:

(A) Ratio
(B) Proportion
(C) Mathematical Biology
(D) Law

Q6. These are involved in classification, except:

(A) Analogous
(B) Homologous
(C) Cytology
(D) Genetics

Q7. Cell organelle found in animal cell and help in intracellular : digestion:

(A) Lysosomes
(B) Ribosomes
(C) Mitochondria
(D) Golgi apparatus

Q8. The comparison of two values is called:

(A) Ratio
(B) Proportion
(C) Graph
(D) Table

Q9. Members of this kingdom have cell wall and are all : heterotrophic:

(A) Monera
(B) Protista
(C) Plantae
(D) Fungi

Q10. Sequence of physiological changes by which cell commit suicide, : is called:

(A) Apoptosis
(B) Necrosis
(C) Apomixes
(D) None of these

Q11. The point where the enzyme is most active, is known as:

(A) Neutral pH
(B) Acidic pH
(C) pH
(D) pH

Q12. Stage of aerobic respiration takes place at the cristae of mitochondria, is called:

(A) Electron transport chain
(B) Glycolysis
(C) Krebs's cycle
(D) C3 cycle C3

Q13. Phenomenon in which pair of homologous chromosome fails to separate is called:

(A) Non disjunction
(B) Terminalization
(C) Synapsis
(D) Linkage

Q14. Each mole of ATP stores energy: : ATP

(A) 7.3 kcal
(B) 7.3 kj
(C) 7.3 kj
(D) 7.3 kcal

Q15. Chemical reaction requires these particular conditions to carry down at Proper rate:

(A) Temperature and nature
(B) Nature and pressure
(C) Nature and structure
(D) Temperature and pressure

Q16. Deficiency of vitamin – K causes the disease of: : K

(A) Rickets
(B) Anemia
(C) Scurvey
(D) Beriberi

Q17. Movement of Water molecules from the area of higher : potential to the area of lower water potential, is called:

(A) Diffusion
(B) Osmosis
(C) Active transport
(D) Water potential

Q18. Secondary cell wall is made up of:

(A) Pectin and Cellulose
(B) Cellulose and Protein
(C) Cellulose and Lignin
(D) Lignin and Pectin

Q19. All factors fled enzyme's activity, except:

(A) substrate concentration
(B) Organic solvent
(C) Temperature
(D) pH

Q20. These are involved in the clotting of blood:

(A) Platelets
(B) Haemoglobin
(C) Albumin
(D) Globulin

Q21. The movement of organic material (food) takes place through: : ( )

(A) Xylem
(B) Vessels
(C) Tracheid
(D) Phloem

Q22. In human digestive system, trachea and esophagus both : connect to the:

(A) Large intestine
(B) Stomach
(C) Pharynx
(D) Rectum

Q23. Splitting of water in presence of light, is called:

(A) Hydrolysis
(B) Glycolysis
(C) Photolysis
(D) None of these

Q24. Metabolic reactions are:

(A) Constructive
(B) Destructive
(C) Inhibiting
(D) None of these

SECTION “B” : (Short Answer Question)

Maximum Marks: 24

Note: Note: Attempt any EIGHT questions from this section. Each question carries Three (3) marks.

Q2. Define: Paleontology, Environmental biology, Genetics. :

Q3. Draw a neat and labelled diagram of Animal cell.

Q4. Define classification. Names units of it and explain any one.

Q5. Describe the relationship of Biology with any three branches of science.

Q6. Define plastids and write two types of it.

Q7. Write three differences between prokaryotic cell and eukaryotic cell.

Q8. Define chromosome. Name the types of it.

Q9. Define enzyme. Write any two uses of enzymes.

Q10. Write three differences between light reaction and dark reaction

Q11. Define heterotrophic nutrition and explain its types.

Q12. Define vitamin and explain its two types.

Q13. Write any three effects of malnutrition on human.

SECTION “C” : (Detailed –Answer Question)

Maximum Marks: 12

> Note: Note: Attempt any Two questions from this Section. Each question carries SIX (6) marks.

Q14. Draw a neat and labelled diagram of human alimentary canal.

Q15. Define transpiration and explain factors which affect its rate.

Q16. Define respiration. Describe the usage of respiration energy in the body of organism. ----OR---- Define tissue. Explain muscle tissue and its types in detail.